Separating Consequence and Tone in Preference Feedback

T

Abstract

Human feedback can reward both relief of a problem and the feeling of having been heard. We specify a factorial feedback design that estimates both contributions and exposes a ranking reversal. Two structural models with identical observed training examples have different causal effects. The result motivates interventional examples during reward-model development and an explicit outcome holdout. Three fitted affine scorers illustrate the role of the training target. An outcome target chose benefit in all 36 synthetic holdout pairs. The pooled rating target chose benefit in twelve and tied in one. Two opposite-corner comparisons identify average effects but can leave the effect of helping ambiguous at a fixed tone. We derive sharp bounds for that effect from the two comparisons. When the comparisons are reported as intervals, a graph gives exact effect bounds and detects incompatible panels.

Summary and scope

Question. What must a preference study vary to separate genuine help from a tone cue?

Finding. Opposite-corner comparisons identify average effects. The missing factorial interaction can still reverse the consequence effect at one tone. Four independently defined cells resolve that ambiguity in the rating table. An interval-valued panel can sometimes certify the sign using a single extra comparison.

Evidence and scope. The bounds are exact and checked by enumeration. The training results use synthetic consequence labels and supplied features. They characterize those learning targets and leave the validity of real outcome labels to separate adjudication.

1 What a high rating buys

A warm dismissal can receive a higher immediate rating than a plain answer that solves the problem. Comfort can itself be valuable. The development risk is teaching a tone cue to substitute for the sought help. Constitutional AI demonstrates a training pipeline using critiques, revision and preference feedback [1]. Reward-gaming analysis makes the general proxy problem precise [2]. Here the intervention changes the example design upstream of that proxy.

Let h be a declared helpful consequence and t a tone treatment, both binary. In a synthetic crossed panel take rating $R(h, t) = 2h + 3t$. At fixed tone, helping raises the score by two in both cells. Yet $R(0, 1) = 3 > 2 = R(1, 0)$. A scalar fitted perfectly to this rating can reproduce that reversal. Crossed grading makes the tradeoff visible even when the fit looks excellent.

2 What two paired comparisons can establish

Suppose the assessor keeps only two comparisons. In the first, the useful response is warm and the unhelpful one is plain. In the second, those tone assignments are reversed. Write $R_{ht} \in [0,1]$ for the normalized mean rating, with $h = +$ denoting the useful response and $t = +$ the warm tone. The observed contrasts are

$$A = R_{++} - R_{--}, \qquad B = R_{+-} - R_{-+}.$$

Their half-sum is the average effect of the useful response across tones, and their half-difference is the average tone effect across consequences. Neither contrast gives a direct comparison of consequences at a fixed tone.

Proposition 1. *Let $W = 2 - |A| - |B|$. With only these two contrasts and the unit rating range, the sharp interval for the factorial interaction $C = R_{++} + R_{--} - R_{+-} - R_{-+}$ is $[-W, W]$. The two fixed-tone consequence effects are $(A + B + C)/2$ and $(A + B - C)/2$. Each has sharp interval*

$$\left[\frac{A + B - W}{2}, \frac{A + B + W}{2}\right].$$

Proof. Two numbers in $[0,1]$ whose difference is A can have any sum in $[|A|, 2 - |A|]$. The other pair can independently have any sum in $[|B|, 2 - |B|]$. Taking the difference of these sums gives the interaction interval, with both endpoints attainable. Expanding each fixed-tone contrast in terms of A, B, C yields the claimed effects and endpoints. □

For example, take $(A, B) = (0.6, 0.2)$. The average consequence effect is 0.4, while the fixed-tone effect can be -0.2. In order $(++, +-, -+, --)$, the table $(0.6, 1, 0.8, 0)$ gives effects -0.2 and 1. The table $(0.8, 0.6, 0.4, 0.2)$ gives 0.4 at both tones with the same observed contrasts. Measuring one fixed-tone comparison in the same rating population identifies the missing interaction. If the claims concern actual user benefit, the outcome check still needs independent validation.

Across 17,667 four-cell tables on three rational grids, exact enumeration recovered the predicted extrema for every contrast pair. This finite check complements the continuous proof and does not estimate how often the ambiguity occurs in human feedback.

3 Combining comparisons reported as intervals

Different reviewers may estimate different contrasts. Suppose each reported interval bounds the difference of two cells of one declared reference population. The intervals must already allow for sampling uncertainty and any population differences the audit permits. We ask which rating tables remain consistent with all of them.

Include a vertex for each cell and an anchor vertex with score zero. For a bound $\ell \le R_i - R_j \le u$, add an edge $j \to i$ of weight u and an edge $i \to j$ of weight $-\ell$. Connect the anchor to each cell and back with weights one and zero, respectively. Write $d(i, j)$ for the shortest path length from i to j.

Proposition 2. *A common rating table exists exactly when the graph has no negative cycle. In that case the sharp interval for $R_a - R_b$ is $[-d(a, b), d(b, a)]$.*

Proof. Each edge bounds a difference of scores. Adding the constraints around a cycle makes its weight nonnegative. Conversely, if there is no negative cycle, distances from any one vertex are

finite because all cells connect to the anchor. The triangle inequality makes those distances a feasible set of scores after subtracting the anchor distance. Adding the inequalities along any path gives the stated interval. Using distances from b attains the upper endpoint, and using distances from a attains the lower endpoint. The feasible tables form a convex set, so every intermediate value is also attained. □

For example, suppose the diagonals give $A \in [0.55, 0.65]$ and $B \in [0.15, 0.25]$. The warm-tone consequence effect could be as low as -0.30 or as high as 1. A third comparison that puts the plain-tone effect in $[0.50, 0.60]$ narrows the warm-tone interval to $[0.10, 0.40]$. The sign is now determined even though none of the three comparisons is exact. An audit can use this test to decide whether paying for a fourth comparison could change the sign conclusion under the declared intervals.

There is also a direct inconsistency check. With diagonals 0.60 and 0.20, the two fixed-tone effects must sum to 0.80. Intervals $[0.30, 0.50]$ and $[0.70, 0.90]$ each fit the rating scale but have a combined lower bound of 1. A cycle detects the conflict. The diagnosis is incompatibility under the common-table assumption. It cannot identify which panel, uncertainty interval, or population assumption failed.

An exhaustive oracle classified 256 finite panels, of which 66 were incompatible. On the 190 feasible panels it confirmed all 4,560 lower and upper endpoints across ordered cell pairs. With integer-scaled constraints the distance witnesses are integer-scaled too, so the grid oracle covers the continuous extrema in these cases.

4 An observational obstruction

Proposition 3. *There are two causal models with the same observed distribution of action and outcome and different average effects of action.*

Proof. Draw X uniformly from $\{0, 1\}$ and set $A = X$. In model I let $Y = A$. In model II let $Y = X$. Both observational laws put half their mass on each of $(0, 0)$ and $(1, 1)$. Replacing the action equation by $A = a$ sets $Y = a$ in I and leaves its mean $1/2$ in II. The average effects are one and zero, respectively. This uses the structural intervention semantics described by Pearl [3]. □

5 Development protocol

Build four matched alternatives per scenario wherever the variation is legitimate. Hold intent, privacy and permission constant. Independently vary tone and the concrete effect, such as whether a correction actually fixes the error. Record the beneficiary's stated goal and an outcome check separately from satisfaction. A warm explanation can win within the set that meets the task and respects the beneficiary. In cases where tone changes uptake, keep the interaction term $R(1, 1) - R(1, 0) - R(0, 1) + R(0, 0)$ rather than forcing separability.

Compare an ordinary preference reward against a reward with separately supervised outcome and tone heads. Freeze scenario families and check whether the latter reduces warm-but-unhelpful choices when the two signals conflict. Include neutral-but-helpful and harmful-but-popular alternatives, with independent outcome adjudication. Train only in settings where synthetic or consented outcomes can be checked. Wrong outcome labels remain dangerous under either design.

6 Audit and next measurement

The local check enumerated all four rating cells, confirmed contrasts $(2, 2)$, and evaluated both intervention distributions. The synthetic effect magnitudes are a counterexample to identification from passive ratings.

A separate training job fitted three affine scorers on the same four balanced feature rows from zero initialization. Each used eight exact gradient steps of size $1/2$. The feature columns are orthogonal and have unit mean square. A closed-form least-squares oracle therefore checks each fitted vector as $255/256$ times its optimum. The targets were pooled rating, direct helpful outcome, and deliberately reversed outcome. The 36 held-out pairs crossed new strength values for both features. Each compares a helpful plain action with a warm unhelpful one.

Training target	Chose help	Tied	Chose warm unhelpful
Pooled rating	12	1	23
Direct outcome	36	0	0
Reversed outcome	0	0	36

The reversed target is the decisive source-validity check. The synthetic model learns the goal developers specify, including a wrong one. Here legitimate and proxy features are supplied by the experimenter. A future model trial needs outcome fidelity and realistic crossing. A manipulation that changes the apparent tone and what was actually offered must be treated as a different intervention.

7 Matched stochastic training study

Eight train and holdout seed pairs tested whether the earlier result survives noisy supervision. We fitted logistic scores from 320 balanced-in-expectation feature draws per seed. The pooled labels were Bernoulli with logit $0.9h + 2t$, and direct labels had logit $2h$. The two arms used the same inputs, uniforms, optimizer, and evaluation pairs. Each new pair compared a helpful colder choice with a warm unhelpful choice at unseen feature strengths.

The direct outcome arm chose help on 3,200 of 3,200 pairs, compared with 58 for pooled feedback. The paired gain ranged from 94.75 to 100 percentage points across the eight seeds. This extends the deterministic example to noisy labels within the supplied feature map. The latent target remains an experimenter definition.

Review status. Independent manuscript review and a language-model study remain open.

References

[1] Yuntao Bai et al. *Constitutional AI: Harmlessness from AI Feedback*. `https://arxiv.org/pdf/2212.08073`.

[2] Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. *Defining and Characterizing Reward Hacking*. `https://arxiv.org/pdf/2209.13085`.

[3] J. Pearl. Causal inference in statistics: An overview. `https://ftp.cs.ucla.edu/pub/stat_ser/r350.pdf`.